

Assignment – 1

Student Name: Yashi Shakya

Enrollment Number: 23301012024

College Name: IGDTUW

1 Find area of rectangle.

```
# Program to find area of rectangle

length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))

area = length * breadth

print("Area of Rectangle =", area)
```

```
... Enter length: 5
Enter breadth: 4
Area of Rectangle = 20.0
```

2 Find simple interest.

```
# Program to calculate simple interest

principal = float(input("Enter principal amount: "))
rate = float(input("Enter rate of interest: "))
time = float(input("Enter time (in years): "))

simple_interest = (principal * rate * time) / 100

print("Simple Interest =", simple_interest)
```

```
... Enter principal amount: 200
Enter rate of interest: 10
Enter time (in years): 3
Simple Interest = 60.0
```

3 Convert temperature from Celsius to Fahrenheit.

```
# Program to convert Celsius into Fahrenheit

celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9/5) + 32

print("Temperature in Fahrenheit =", fahrenheit)
```

```
... Enter temperature in Celsius: 35
Temperature in Fahrenheit = 95.0
```

4 Calculate average of 3 numbers.

```
# Program to find average of three numbers

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))

average = (num1 + num2 + num3) / 3

print("Average =", average)
```

```
... Enter first number: 3
Enter second number: 5
Enter third number: 6
Average = 4.666666666666667
```

5 Find square and cube of a number.

```
# Program to find square and cube

number = int(input("Enter a number: "))

square = number ** 2
cube = number ** 3

print("Square =", square)
print("Cube =", cube)
```

```
... Enter a number: 3
Square = 9
Cube = 27
```

6 Swap two numbers without third variable.

```
# Program to swap two numbers without third variable

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

a = a + b
b = a - b
a = a - b

print("After swapping:")
print("First number =", a)
print("Second number =", b)
```

```
... Enter first number: 2
Enter second number: 5
After swapping:
First number = 5
Second number = 2
```

7 Create a Student Report Program that take student details using input(), Store marks in variables, Calculate total and percentage , Use comments, Use proper indentation

```
▶ # Student Report Program
name = input("Enter student name: ")
roll_no = input("Enter roll number: ")

# Taking marks of three subjects
maths = float(input("Enter Maths marks: "))
science = float(input("Enter Science marks: "))
english = float(input("Enter English marks: "))
# Calculating total marks
total = maths + science + english

# Calculating percentage
percentage = total / 3
# Displaying report
print("\n----- STUDENT REPORT -----")
print("Student Name :", name)
print("Roll Number :", roll_no)
print("Maths Marks :", maths)
print("Science Marks:", science)
print("English Marks:", english)
print("Total Marks :", total)
print("Percentage :", percentage, "%")
```

```
... Enter student name: yashi
Enter roll number: 233
Enter Maths marks: 96
Enter Science marks: 97
Enter English marks: 94

----- STUDENT REPORT -----
Student Name : yashi
Roll Number : 233
Maths Marks : 96.0
Science Marks: 97.0
English Marks: 94.0
Total Marks : 287.0
Percentage : 95.66666666666667 %
```